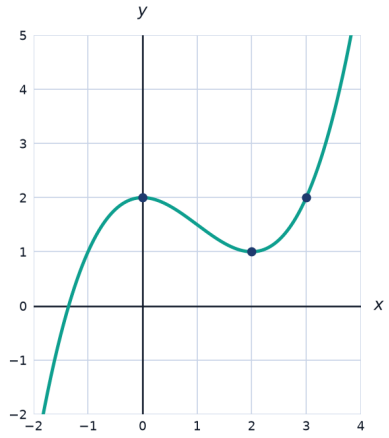
# Factorising non-monic trinomials (Extended)

## Core Concepts: Factorising non-monic trinomials (Extended)

**1** For the model $f(x) = 5x + 6$ in factorising non-monic trinomials (extended), evaluate $f(2)$.




**a.** 16

**b.** 21

**c.** 10

**d.** 20

**2** Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

**a.** $x = 1$ and $x = 7$

**b.** $x = 1$ and $x = 6$

**c.** $x = -1$ and $x = -6$

**d.** $x = 7$ and $x = 6$

**3** A data set recorded for factorising non-monic trinomials (extended) consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 5x + 6$ in factorising non-monic trinomials (extended), evaluate $f(3)$.

**a.** 21

**b.** 26

**c.** 15

**d.** 25

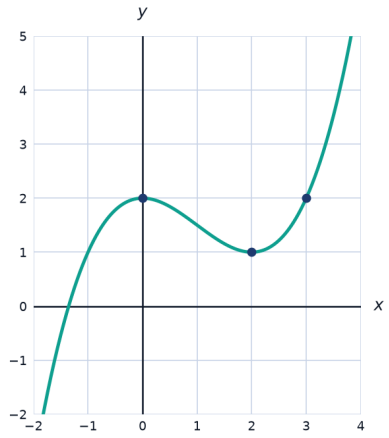
**5** Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

**a.** $x = 2$ and $x = 8$

**b.** $x = 2$ and $x = 7$

**c.** $x = -2$ and $x = -7$

**d.** $x = 9$ and $x = 14$




**6** A data set recorded for factorising non-monic trinomials (extended) consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 5x + 6$ in factorising non-monic trinomials (extended), evaluate $f(4)$.

**a.** 26

**b.** 31

**c.** 20

**d.** 30

**8** Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ and $x = 8$

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

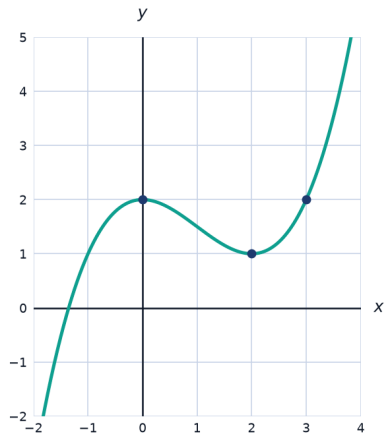
**9** A data set recorded for factorising non-monic trinomials (extended) consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

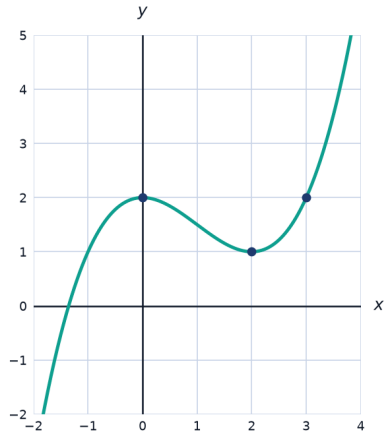
**d.** 26.5




## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 5x + 6$ in factorising non-monic trinomials (extended), evaluate $f(5)$.




**a.** 31

**b.** 36

**c.** 25

**d.** 35

**11** Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

**a.** $x = 4$ and $x = 10$

**b.** $x = 4$ and $x = 9$

**c.** $x = -4$ and $x = -9$

**d.** $x = 13$ and $x = 36$

**12** A data set recorded for factorising non-monic trinomials (extended) consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

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